



Project Status report

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Community & UN SDG(s): **SaskTel network engineers and architects**

UN SDG(s):

- SDG#7: Affordable and clean energy
- SDG#11: Sustainable cities and communities
- SDG#12: Responsible consumption and production
- SDG#13: Climate action

MVP # **1-4**

Sprint cycle dates: **March 25th, 2025 – April 7th, 2025**

Project Name	Eco-Resilient Networks: Smart Deployment for the Future
Blurb	We have successfully developed Eco-Resilient Networks: Smart Deployment for the Future , a simulation framework that optimizes Service Function Chain (SFC) deployment in edge-cloud environments for 5G video streaming. By jointly maximizing network availability and minimizing carbon footprint , our solution empowers SaskTel network engineers and architects to make data-driven decisions, directly addressing UN Sustainable Development Goals 7 (Affordable and Clean Energy), 11 (Sustainable Cities), 12 (Responsible Consumption), and 13 (Climate Action) .
For Week Ending	April 7 th , 2025
Project Status	Green
Status Description	The project has been successfully executed according to the planned schedule, with all Minimum Viable Products (MVPs) delivered and final deliverables prepared for submission. Key Accomplishments: MVP 1 (Simulation Environment Setup): ○ Fully implemented core framework (fat-tree topology, SFC generation, VNF placement). ○ Completed data I/O modules (JSON/CSV integration) and visualization tools (Matplotlib). MVP 2 (Algorithm Implementation & Evaluation): ○ Embedding Policies: Availability-Aware (AAE), Carbon-Aware (CAE), and Tradeoff-Aware (TAE) policies integrated and tested. ○ Optimization Algorithms: Particle Swarm Optimization (PSO), Simulated Annealing (SA), and Genetic Algorithm (GA) implemented and benchmarked. ○ Performance Metrics: Validated against SLA constraints (e.g., >99.9% availability, <100ms latency). MVP 3 (Documentation & Reporting): ○ User Guide: Step-by-step instructions for installation, configuration, and simulation. ○ Research Paper: Finalized with methodology, results, and discussion sections. ○ Code Documentation: Docstrings and comments for maintainability. MVP 4 (Delivery): ○ Integrated all components into a functional standalone Python application. ○ Prepared a 1–2 minute video commercial (non-technical demo) and technical presentation (15-minute overview + Q/A).

Activities—During the past sprint cycle



During the past sprint cycle, we achieved the following:

1. **Work Package 2.5: Other Redundancy Optimization Algorithms (Finalized)**
 - Completed implementation of **Simulated Annealing (SA)** and **Genetic Algorithm (GA)**.
 - Integrated SA/GA with embedding policies (AAE, CAE, TAE) for benchmarking.
 - Generated final performance comparisons across all algorithms (PSO/SA/GA).
2. **Work Package 3.1: Code Documentation (Completed)**
 - Finalized docstrings, comments, and repository README.
 - Ensured modular code structure.
3. **Work Package 3.2: User Guide (Completed)**
 - Completed all sections:
 - Installation (Python and dependencies setup).
 - Configuration (topology/SFC input files).
 - Running simulations (policy/algorithm selection).
 - Interpreting results (metrics, visualizations).
4. **Work Package 3.3: Research Paper (Completed)**
 - Finished **Results** section (statistical analysis, trade-off graphs).
 - Finalized **Discussion** (algorithm performance insights) and **Conclusion** (SDG impact, future work).
5. **Work Package 4.1: MVP Integration & Testing**
 - Merged all components into a standalone Python application.
 - Validated end-to-end workflow with edge cases.
6. **Work Package 4.2: MVP Demo & Presentation**
 - Recorded **video commercial** (1–2 min) highlighting SDG alignment and key features.
 - Prepared **technical presentation** (15 min) covering:
 - SDG context and gap analysis.
 - Live demo of TAE + PSO optimization.
 - Reflection on lessons learned.

Project Issues

No significant issues arose during this final sprint that impacted project delivery or required escalation. All minor technical challenges encountered were resolved within the sprint through team collaboration.

Project Changes

No changes to project scope, requirements, or deliverables were requested or approved during this final sprint. The project was executed as originally defined in:

- Project Scope Statement (P06)
- Project Requirements (P07)
- Activity-Based Schedule (P08)

Activities—Planned for Next Week

With all deliverables submitted and the project formally concluded, the team will:

1. **Present Final Work:**
 - Deliver the 15-minute technical presentation and Q/A session as scheduled.
 - Share the video commercial with stakeholders (e.g., SaskTel, RCE Saskatchewan).
2. **Archive Materials:**
 - Finalize GitHub repository (e.g., add release tags, clean up branches).
 - Backup all project assets (code, datasets, documentation).
3. **Reflect & Transition:**



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- Submit peer feedback on colleagues' projects.
 - Document lessons learned for future research or professional development.

Reflection

Do you feel "on track"?

Yes, the project was completed on schedule with all deliverables meeting or exceeding expectations. The final MVPs align perfectly with our initial goals of optimizing SFC deployment for both availability and sustainability.

What progress do you particularly feel good (great) about?

1. **Algorithm Innovation:** Successfully implementing and benchmarking three embedding policies (AAE, CAE, TAE) with PSO/SA/GA, demonstrating clear trade-offs between availability and carbon footprint.
2. **SDG Impact:** Directly linking technical outcomes to UN SDGs—especially reducing energy use (SDG 7) while maintaining resilient infrastructure (SDG 9).
3. **Professional Deliverables:** Producing a publishable research paper and user-friendly documentation, ensuring the work's longevity beyond the course.

What barriers (if any) do you feel is/are a current impediment to success?

No significant barriers were encountered throughout the project.

What help (if any) do you require to move positively forward?

- **Industry Feedback:** Input from SaskTel engineers on real-world applicability would validate next steps.
- **Research Collaboration:** Partnering with sustainability researchers to deepen carbon footprint modelling.

What questions or concerns do you have (if any)?

- How might latency constraints evolve with 6G, and how should algorithms adapt?
 - Could this framework integrate with SaskTel's existing SDN controllers long-term?
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